Conduction, convection, radiation

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Learning target: Identify mechanisms of heat transfer and support each classification.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 22.*
5 W then 2.5 W. Same energy over twice the time gives half the power.

02. Worked example: explain the model
Conduction via microscopic interactions; convection via bulk fluid motion; radiation via electromagnetic waves.

03. Read and interpret the model
Conduction; convection; radiation. Accept noting multiple effects if primary mechanism is justified.

04. Apply the physics
Bulk fluid movement versus microscopic transfer in a solid; buoyancy-driven loop in stated heating arrangement.

05. Explain your reasoning
Radiation; no surrounding air for convection in the near-vacuum model.

06. Review: connect an earlier idea *Review: Unit 4, Day 13.*
Difference 8 J may be rotational energy and thermal energy; other justified transfers acceptable.

07. Exit check
Electromagnetic radiation crosses vacuum.

08. STRETCH: test the model
Examples low-conductivity wall, lid limiting fluid circulation/evaporation, reflective surface reducing radiative transfer; explain limits.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 22.*
v_{escape}=10 m/s. Assumptions include ideal central gravity, no atmosphere, no further thrust and negligible other bodies.

02. Worked example: explain the model
Conduction via microscopic interactions; convection via bulk fluid motion; radiation via electromagnetic waves.

03. Read and interpret the model
Conduction; convection; radiation. Accept noting multiple effects if primary mechanism is justified.

04. Apply the physics
12,000 J; a larger thickness lengthens the conduction path; ideal steady conduction rate inversely proportional to thickness.

05. Explain your reasoning
Radiation; no surrounding air for convection in the near-vacuum model.

06. Review: connect an earlier idea *Review: Unit 4, Day 13.*
mgh≈2.352 J; K=0.8 J; fraction≈0.3401 (34.01%).

07. Exit check
Net thermal radiation; any cooler radiative environment can receive net energy, not only 0 K.

08. STRETCH: test the model
Conduction through supports/neck and radiation remain; compare neck insulation and surface emissivity while controlling temperatures. Vacuum suppresses fluid conduction/convection, not all transfer.